

first name

last name

No Calculators.

Questions 1 and 2. Circle **True** or **False**. Then, in **one sentence**, say why — if the statement is false, the fastest way is usually to give an example that breaks it. **1 point for the circle, 2 for the reason.**

1. (3 pts) The average rate of change of f on $[a, b]$ is the slope of the secant line through $(a, f(a))$ and $(b, f(b))$.

True **False**

/3

Why? _____

2. (3 pts) If $\lim_{x \rightarrow 2} f(x) = 5$, then $f(2) = 5$.

True **False**

/3

Why? _____

3. (4 pts) Let $f(x) = 3 - x^2$. Form the difference quotient at $a = 2$ and simplify it as far as it will go:

$$\frac{f(2+h) - f(2)}{h}$$

/4

4. (4 pts) A drone rises straight up. Its height above the ground after t seconds is

$$s(t) = 40t - 5t^2 \text{ metres, } 0 \leq t \leq 8.$$

/4

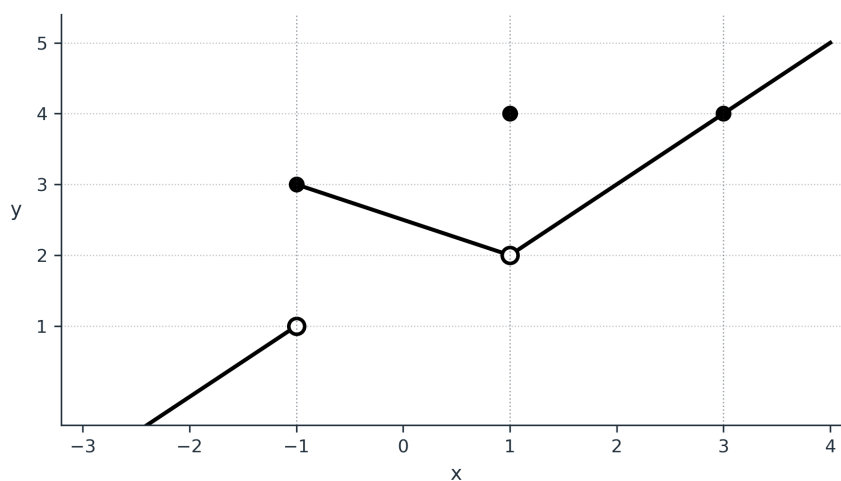
- (a) Find the drone's **average velocity** over the interval $[1, 3]$. Include units.

- (b) In **one sentence**, say what that number tells you about the drone.

- (c) Over $[1, 1.5]$, $[1, 1.1]$ and $[1, 1.01]$ the average velocities are 27.5, 29.5 and 29.95 m/sec. What number are they heading for, and what does it say about the drone at the **instant** $t = 1$?

5. (6 pts) Use the graph of $y = f(x)$ below. If a limit does not exist, say so.

/6



(a) $\lim_{x \rightarrow -1^-} f(x) =$ _____ (b) $\lim_{x \rightarrow -1} f(x) =$ _____

(c) $\lim_{x \rightarrow 1} f(x) =$ _____ (d) $f(1) =$ _____

(e) $\lim_{x \rightarrow 3} f(x) =$ _____ (f) $f(3) =$ _____